MATH 217: Probability and Statistics Özyegin University

Recitation III

Section: A

Problem 1.

The amplitude V (volts) of a sinusoidal signal is a discrete uniform (-3,3) random variable. Let $Y = V^2/2$ watts denote the power of the transmitted signal. Find $P_Y(y)$.

Solution 1.

The possible values of Y are $S_Y = \{0, 0.5, 2, 4.5\}$. Since $Y = y$ when $V = \sqrt{2y}$ or $V = -\sqrt{2y}$, we see that

$$P_Y(0) = P_V(0) = 1/7.$$

For $y = 0.5, 2, 4.5$,

$$P_Y(y) = P_V(\sqrt{2y}) + P_V(-\sqrt{2y}) = 2/7.$$

Therefore,

$$P_Y(y) = \begin{cases} 1/7 & y = 0, \\ 2/7 & y = 0.5, 2, 4.5, \\ 0 & otherwise \end{cases}$$

Problem 2.

In an experiment to monitor two packets, the PMF of N, the number of video packets, is

n	0	1	2
$P_N(n)$	0.2	0.7	0.1

Find $E[N]$, $E[N^2]$, Var[N], and σ_N.

Solution 2.

Given the following PMF

$$P_N(n) = \begin{cases} 0.2 & n = 0, \\ 0.7 & n = 1, \\ 0.1 & n = 2, \\ 0 & otherwise. \end{cases}$$

the calculations are straightforward:

a. E[N] = (0.2)0+(0.7)1+(0.1)2 = 0.9.

b. $E[N^2] = (0.2)0^2 + (0.7)1^2 + (0.1)2^2 = 1.1$.

c. $Var[N] = E[N^2] - E[N]^2 = 1.1 - (0.9)^2 = 0.29$.

d. $\sigma = \sqrt{Var[N]} = \sqrt{0.29}$.

Problem 3.

Let X have the binomial PMF

$$P_X(x) = \binom{4}{x}(1/2)^4$$

a. Find the standard deviation of X.

b. Find $P[\mu_X - \sigma_X \leq X \leq \mu_X + \sigma_X]$, the probability that X is within one standard deviation of the expected value.

Solution 3.

a. The expected value of X is

$$\begin{aligned} E[X] &= \sum_{x=0}^{4} x P_X(x) \\ &= 0\binom{4}{0}\frac{1}{2^4} + 1\binom{4}{1}\frac{1}{2^4} + 2\binom{4}{2}\frac{1}{2^4} + 3\binom{4}{3}\frac{1}{2^4} + 4\binom{4}{4}\frac{1}{2^4} \\ &= [4 + 12 + 12 + 4]/2^4 = 2. \end{aligned}$$

The expected value of X^2 is

$$\begin{aligned} E[X^2] &= \sum_{x=0}^{4} x^2 P_X(x) \\ &= 0^2\binom{4}{0}\frac{1}{2^4} + 1^2\binom{4}{1}\frac{1}{2^4} + 2^2\binom{4}{2}\frac{1}{2^4} + 3^2\binom{4}{3}\frac{1}{2^4} + 4^2\binom{4}{2}\frac{1}{2^4} \\ &= [4 + 24 + 36 + 16]/2^4 = 5. \end{aligned}$$

The variance of X is

$$Var[X] = E[X^2] - (E[X])^2 = 5 - 2^2 = 1;$$

Thus, X has standard deviation $\sigma_X = \sqrt{Var[X]} = 1$.

b. The probability that X is within one standard deviation of its expected value is

$$\begin{aligned} P[\mu_X - \sigma_X \leq X \leq \mu_X + \sigma_X] &= P[2 - 1 \leq X \leq 2 + 1] \\ &= P[1 \leq X \leq 3]. \end{aligned}$$

This calculation is easy using the PMF of X :

$$P[1 \leq X \leq 3] = P_X(1) + P_X(2) + P_X(3) = 7/8.$$

Problem 4.

Seven balls are randomly withdrawn from an urn that contains 12 red, 16 blue and 18 green balls. Find the probability that

(a) 3 red, 2 blue and 2 green balls are withdrawn; (b) either exactly 3 red balls or exactly 3 blue balls are withdrawn.

Solution 4.

Problem 5.

(a) $$\frac{\binom{12}{3}\binom{16}{2}\binom{18}{2}}{\binom{46}{7}}$$

(b) $$1-\frac{\binom{34}{7}}{\binom{46}{7}}-\frac{\binom{12}{1}\binom{34}{6}}{\binom{46}{7}}$$

(c) $$\frac{\binom{12}{7}+\binom{16}{7}+\binom{18}{7}}{\binom{46}{7}}$$

Consider an unending sequence of independent trials where each trial is equally likely to result in any of the outcomes 1, 2, or 3. Given that outcome 3 is the last of the three outcomes to occur, find the conditional probability that

(a) the first trial results in outcome 1; (b) the first two trials both result in outcome 1

Solution 5.

(a) This is equal to the conditional probability that the first trial results in outcome 1 (F_1) given that it results in either 1 or 2, giving the result 1/2. More formally, with L_3 being the event that outcome 3 is the last to occur

$$P(F_1 \mid L_3) = \frac{P(L_3 \mid F_1)\,P(F_1)}{P(L_3)} = \frac{(1/2)(1/3)}{1/3} = 1/2$$

(b) With S_1 being the event that the second trial results in outcome 1, we have

$$P(F_1S_1 \mid L_3) = \frac{P(L_3 \mid F_1S_1)\,P(F_1S_1)}{P(L_3)} = \frac{(1/2)(1/9)}{1/3} = 1/6$$

Problem 6.

If there are 12 strangers in a room, what is the probability that no two of them celebrate their birthday in the same month?

Solution 6.

$12!/12^{12}$

Problem 7.

The probability that a student pilot passes the written test for a private pilot's license is 0.7. Find the probability that a given student will pass the test

(a) on the third try;

(b) before the fourth try.

Solution 7.

Using the geometric distribution

(a) $P(X = 3) = g(3; 0.7) = (0.7)(0.3)^2 = 0.0630.$

(b) $P(X < 4) = \sum_{x=1}^{3} g(x; 0.7) = \sum_{x=1}^{3} (0.7)(0.3)^{x-1} = 0.9730.$